

PAIMANA Data Acquisition & Verification Report

SIH 2026 — Problem Statement 26103 (MoSPI, IPMD) · generated 30 August 2026, 23:03

How to read this report

Every figure is read from a data file at generation time by `scripts/build_report.py`. No value is hand-entered. Where a source file is absent or a run is unfinished, the report states that instead of estimating. Regenerate at any time to get current numbers.

1. What was acquired

Two independent sources were harvested. They are complementary: the dashboard API gives rich fields over a short window, the PDF archive gives a long history at lower field richness.

source	volume	size	coverage
Monthly Flash Reports (PDF)	207 files	1.91 GB	2001–2026
Quarterly PISR (PDF)	7 files	included above	—
Dashboard API panel (JSON)	18,601 rows	2,243 projects	2025-07 – 2026-07

Coverage of the monthly archive

year	J F M A M J J A S O N D	n
2001	...#####	9/12
2002	#####	12/12
2003	###.....	3/12
2004		0/12
2005		0/12
2006		0/12
2007	...#. ...#..	2/12
2008		0/12
2009	..#..#####	8/12
2010	#####	12/12
2011	#####	12/12
2012	#####	12/12
2013	..#####	11/12
2014	#####	12/12
2015	#####	12/12
2016	#####	12/12
2017	#####	12/12
2018	#####.....	7/12
2019		0/12
2020	..####. #####	10/12
2021	#####	12/12
2022	#####	12/12
2023	#####	12/12
2024	#####. #####	11/12
2025	#####	12/12

year	J F M A M J J A S O N D	n
2026	#. #	2/12

2. Verification

Every file was opened and checked for a valid PDF header, an extractable text layer, and the presence of project rows.

check	result
Files carrying project-level data	167 of 197
Files without project rows	30
Total pages verified	65,733

The excluded files are not corrupt

They are short executive-summary editions (17–26 pages) with a perfectly good text layer that simply do not contain the project annexures. Consequence: the usable project-level panel begins in 2009, not 2001. A panel built naively across all files would show those months as empty.

3. Measured findings — dashboard panel

snapshot	projects	time overrun	cost overrun	median delay (mo)	median cost overrun
2026-04	1,981	77.9%	54.7%	21.3	18.9%
2026-07	1,775	79.1%	56.8%	22.5	19.0%

Overruns are computed from RevisedCost vs OriginalCost and RevisedDate vs OriginalEndDate. The API's own overrun fields (DELAYED_TIME, COST_OVERRUN, COR_PERC, TOR_PERC) are returned 0% populated and cannot be used.

Why the base rate decides everything

With roughly four in five monitored projects already behind schedule, a model that predicts “delayed” for every project scores that rate for free. Any accuracy claim must be stated against this baseline, not against 50%.

4. Structured extraction from the PDF archive

Status: complete

measure	value
Rows extracted	879,643
Project-level rows	592,608 (67.4%)
Aggregate / summary rows (excluded from project counts)	287,035 (32.6%)
Project-level rows carrying an OCMS code	518,551 (87.5%)
Distinct OCMS project codes seen	3,887
Slip events (EVENT_SLIP)	11,029
of which carry a delay magnitude	11,002

Section labels assigned

label	rows	share
ADMIN_NO_MILESTONE	89,916	15.2%
STATE_DELAYED	89,145	15.0%
ADMIN_NO_DOC	82,728	14.0%
STATE_ON_SCHEDULE	71,323	12.0%
LIST_ONGOING	67,977	11.5%

label	rows	share
STATE_COST_OVERRUN	67,599	11.4%
UNCLASSIFIED	42,883	7.2%
STATE_TIME_AND_COST	30,188	5.1%
STATE_EXPENDITURE_OVER	12,167	2.1%
EVENT_SLIP	11,029	1.9%
LIFECYCLE_COMPLETED	10,393	1.8%
MODE_PPP	5,328	0.9%

Label accuracy varies by era — state this, do not hide it

Unclassified rows currently stand at 7.2% of project-level rows. On a single modern report tuned during development the figure was 0.3%; across the whole archive it is higher because table layouts and heading wording drift between eras. The corpus-wide figure is the honest one to quote.

5. Identifier linkage — an unsolved problem

The two sources use unrelated identifier systems. PAIMANA replaced OCMS-2006 and renumbered every project.

	identifier	stable within source?	links to the other?
PDF archive	OCMS code e.g. N24001658	yes, across monthly reports	no — name-match rate 1.1%
Dashboard API	ProjectId e.g. 706718	yes, across freeze months	no

Do not present these as one continuous series

Within each source the join is exact and safe. Between them it is probabilistic record linkage on project names typed by different people across two decades. Treat them as two linked panels, and state the caveat wherever a claim spans both.

6. Errors found and corrected during this work

Recorded because the corrected figures are only trustworthy if the corrections are visible. Each was found by checking a derived number against the source document.

what went wrong	how it showed up	resolution
Section heading carried forward across pages that had none	EVENT_SLIP inflated to 138,128 rows; slip rates of 80–98% per month, which is not credible	Per-table titles; inheritance now requires a matching column signature
Two-line headings read only on the second line	4,943 rows in one report labelled from “Costing Rs. 150 Crore and above”, which carries no section information	Multiple title candidates including pairwise joins
Fallback accepted any long line as a heading	137 invented sections such as “NAGPUR RING ROAD PACKAGE-2”	Return no title rather than guess; let structural inheritance fill it
Regex deal?y matched “dealy” but not “delay”	3,507 rows dropped to UNCLASSIFIED	Corrected pattern; 13 classifier unit tests added
Aggregate summary tables counted as project rows	Row counts overstated	Tables without a project column flagged is_aggregate
Dates harvested as currency values	“03/2021” contributed 3 and 2021 to cost columns	Dates stripped before money extraction

7. Known limitations

limitation	detail
2019 is entirely absent	272–404 filename variants probed across both archive hosts and both financial-year folder conventions. The files are not published.

limitation	detail
2001–2003 unusable for project data	Executive-summary editions only. Usable panel starts 2009.
CUF specification not public	The Common Upload Form referenced in the problem statement is not published. Any field list is inferred from what the monitoring feed exposes, which may be a subset. Say “fields exposed in the public feed”, not “the CUF”.
Free-text reason fields are empty	RevisedCostReason, RevisedDateReason and Remarks are 0% populated in the public feed. Narrative analysis of delay causes is not deliverable from this data.
Onboarding ramp confounds the early panel	Project counts rise steeply across the first snapshots as projects migrate onto PAIMANA. A project appearing is onboarding, not a new project.

8. Reproduction

The corpus is reproducible without redistributing 1.9 GB: `manifest_all.csv` records the source URL of every file.

step	command
Download the archive	<code>python scripts/download_all.py</code>
Recover missing months	<code>python scripts/fill_gaps.py</code>
Verify integrity	<code>python scripts/verify_corpus.py</code>
Extract tables	<code>python scripts/extract_tables.py</code>
Harvest the dashboard panel	<code>python scripts/scrape_dashboard.py</code>
Compute base rates	<code>python scripts/analyze_dashboard_panel.py</code>
Regenerate this report	<code>python scripts/build_report.py</code>

Sources are public Government of India publications. Content owned and maintained by IPMD, MoSPI.